

Experiment 6: Breadth First Search (BFS)

Aim

To implement the Breadth First Search (BFS) algorithm for graph traversal.

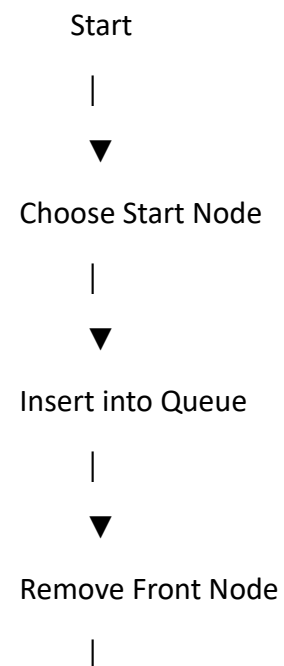
Objective

To traverse all vertices of a graph level by level using a queue.

Algorithm

1. Start from the source vertex.
2. Mark it as visited.
3. Insert it into the queue.
4. Remove the front vertex.
5. Visit all unvisited adjacent vertices.
6. Add them to the queue.
7. Repeat until the queue becomes empty.

Flowchart

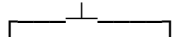




Visit Adjacent Nodes



Queue Empty?



Yes

No



End

Continue

Python Code

```
graph = {  
    'A': ['B', 'C'],  
    'B': ['D', 'E'],  
    'C': ['F'],  
    'D': [],  
    'E': ['F'],  
    'F': []  
}  
  
visited = []  
queue = []  
  
def bfs(node):  
    visited.append(node)  
    queue.append(node)  
  
while queue:
```

```
s = queue.pop(0)
print(s, end=" ")

for neighbour in graph[s]:
    if neighbour not in visited:
        visited.append(neighbour)
        queue.append(neighbour)
```

```
bfs('A')
```

Sample Output

```
A B C D E F
```

Result

Thus, the Breadth First Search (BFS) algorithm was successfully implemented, and all graph nodes were traversed in level order.